

Algebraic Geometry – Final Examination

Solutions

Duration: 1 hour 30 minutes

No documents or electronic devices. All answers must be justified.

The examination contains three problems, each worth 50 points. All points obtained are added, up to a maximum of 100 points. Thus two complete problems are sufficient for full marks; you may work on all three.

Exercise 1. Points over a finite field

For a prime power q , let $\mathbb{F}_q$ denote a field with q elements; in particular, $\mathbb{F}_2 = \mathbb{Z}/2\mathbb{Z}$. Set $X = \operatorname{Spec}(\mathbb{F}_2[t]) = \mathbb{A}_{\mathbb{F}_2}^1$ and $Y = \operatorname{Proj}(\mathbb{F}_2[x_0, x_1]) = \mathbb{P}_{\mathbb{F}_2}^1$.

- (1) Show that $(0), (t), (t+1), (t^3+t+1)$ are points of X . For each point, describe the residue field and determine its cardinality.
- (2) How many $\mathbb{F}_2$ -rational points do X and Y have? Is the set of all points of X finite, countably infinite, or uncountable? Justify your answer.

For $h \in \mathbb{F}_2[t]$, set $Z_h = \operatorname{Spec}(\mathbb{F}_2[t]/(h))$. Let $a \in \mathbb{F}_2$, and set $f = t^2 + at + 1$ and $g = t^3 + at + 1$.

- (3) Determine for which $a \in \mathbb{F}_2$ the scheme Z_f is reduced and when it is connected.
- (4) Show that Z_g is reduced and determine for which $a \in \mathbb{F}_2$ it is irreducible.
- (5) Consider the morphism $F : Y \rightarrow Y$, $[x_0 : x_1] \mapsto [x_0^2 : x_1^2]$. On $D_+(x_0)$, with $t = x_1/x_0$, determine the induced homomorphism $\varphi : \mathbb{F}_2[t] \rightarrow \mathbb{F}_2[t]$. Show that F fixes every point of Y but is not the identity morphism.

Bonus. For the point $p = (t^3 + t + 1)$, describe the induced endomorphism by φ of $\kappa(p)$.

Solution. For (1). The zero ideal is prime because $\mathbb{F}_2[t]$ is a domain. The ideals (t) and $(t+1)$ are maximal. Finally, $t^3 + t + 1$ has no root in $\mathbb{F}_2$; being cubic, it is irreducible, so $(t^3 + t + 1)$ is maximal. Hence all four ideals define points of X . Their residue fields are

$$\kappa((0)) = \mathbb{F}_2(t), \quad \kappa((t)) \simeq \mathbb{F}_2, \quad \kappa((t+1)) \simeq \mathbb{F}_2,$$

and

$$\kappa((t^3 + t + 1)) \simeq \mathbb{F}_2[t]/(t^3 + t + 1) \simeq \mathbb{F}_8.$$

Thus the finite residue fields have cardinalities 2, 2, and 8, while $\mathbb{F}_2(t)$ is countably infinite.

For (2). The $\mathbb{F}_2$ -rational points of X are the closed points of X whose residue field is contained in, hence equal to $\mathbb{F}_2$. A closed point is of the form (p) , where $p \in \mathbb{F}_2[t]$ is irreducible. The residue field $\kappa(p)$ is $\mathbb{F}_2[t]/(p)$, which has cardinality 2^d , where $d = \deg(p)$. Therefore, (p) is $\mathbb{F}_2$ -rational if and only if $d = 1$. In conclusion, the $\mathbb{F}_2$ -rational points of X are (t) and $(t+1)$, so $\#X(\mathbb{F}_2) = 2$.

The projective line has one further rational point, the point at infinity. Indeed, any point of Y lies in $D_+(x_0)$ or in its complement, $\mathbb{W}_+(x_0)$, which is a single point, namely, the point at infinity, that we identify as the $\mathbb{F}_2$ -rational point of the chart $D_+(x_1)$ corresponding to the origin of that chart. So the $\mathbb{F}_2$ -rational points are those of $X \simeq D_+(x_0)$, plus ∞ . In conclusion, $\#Y(\mathbb{F}_2) = 3$.

The points of X are (0) together with the ideals (q) , where q runs over the monic irreducible polynomials in $\mathbb{F}_2[t]$. There are at most countably many because $\mathbb{F}_2[t]$ is countable, and there are infinitely many irreducible polynomials: otherwise, if $q_1, \dots, q_r$ were all of them, any irreducible factor of $q_1 \cdots q_r + 1$ would give another one. Thus X has countably infinitely many points.

For (3). If $a = 0$, then

$$f = t^2 + 1 = (t+1)^2,$$

so Z_f is not reduced. Its underlying topological space has the single point $(t+1)$, hence it is connected. If $a = 1$, then the quadratic polynomial $t^2 + t + 1$ has no root in $\mathbb{F}_2$. It is thus irreducible, so

$$\mathbb{F}_2[t]/(f) \simeq \mathbb{F}_4$$

is a field. Thus Z_f is integral, hence connected, because an affine scheme is integral whenever its ring of regular functions is an integral domain. Consequently Z_f is reduced exactly for $a = 1$, and connected for both values of a .

For (4). If $a = 0$, then

$$g = t^3 + 1 = (t + 1)(t^2 + t + 1).$$

The two factors are distinct and coprime, hence by the Chinese remainder theorem (which applies, since $\mathbb{F}_2[t]$ is a principal ideal domain) we have

$$\mathbb{F}_2[t]/(g) \simeq \mathbb{F}_2 \times \mathbb{F}_4.$$

Thus Z_g is reduced (a product of fields has no non-zero nilpotents) but reducible. If $a = 1$, the cubic polynomial $t^3 + t + 1$ has no root in $\mathbb{F}_2$, hence is irreducible, and the quotient is $\mathbb{F}_8$. Thus Z_g is reduced for both values of a , and irreducible exactly for $a = 1$.

For (5). On $D_+(x_0) \simeq \text{Spec}(\mathbb{F}_2[t])$ the morphism is induced by

$$\varphi : \mathbb{F}_2[t] \longrightarrow \mathbb{F}_2[t], \quad t \longmapsto t^2.$$

For every $h \in \mathbb{F}_2[t]$ one has $h(t^2) = h(t)^2$. Hence, for every prime ideal $\mathfrak{p} \subset \mathbb{F}_2[t]$,

$$h(t^2) \in \mathfrak{p} \iff h(t)^2 \in \mathfrak{p} \iff h(t) \in \mathfrak{p}.$$

Thus every point of this affine chart is fixed. On the other standard chart, with coordinate $s = x_0/x_1$, the same argument gives $s \mapsto s^2$, so every point of Y is fixed. Nevertheless F is not the identity morphism, since its pull-back sends t to $t^2 \neq t$.

Bonus. For $p = (t^3 + t + 1)$ one has $\kappa(p) \simeq \mathbb{F}_8$, and the induced map is

$$\mathbb{F}_8 \longrightarrow \mathbb{F}_8, \quad z \longmapsto z^2.$$

This is the Frobenius automorphism of the field $\mathbb{F}_8$. It is non-trivial; for example, the class of t is sent to the class of t^2 . Actually it generates the Galois group of $\mathbb{F}_8$ over $\mathbb{F}_2$.

Exercise 2. The Nullstellensatz

Let $\mathbb{k}$ be an algebraically closed field and $A = \mathbb{k}[x_1, \dots, x_n]$. You may use without proof *Zariski's lemma*: if a field K is finitely generated as a $\mathbb{k}$ -algebra, then K is a finite algebraic extension of $\mathbb{k}$.

- (1) Let $\mathfrak{m} \subset A$ be maximal. Deduce from Zariski's lemma that there is a unique $a = (a_1, \dots, a_n) \in \mathbb{k}^n$ such that $\mathfrak{m} = (x_1 - a_1, \dots, x_n - a_n)$.
- (2) Deduce that, for every proper ideal $\mathfrak{a} \subsetneq A$, there is $a \in \mathbb{k}^n$ such that $f(a) = 0$ for every $f \in \mathfrak{a}$.
- (3) For an ideal $\mathfrak{a} \subset A$, show that $\sqrt{\mathfrak{a}} \subset \mathbf{I}(\mathbf{V}(\mathfrak{a}))$.
- (4) Let $f \in \mathbf{I}(\mathbf{V}(\mathfrak{a}))$ and set $\mathfrak{b} = (\mathfrak{a}, 1 - yf) \subset A[y]$. Show that $\mathbf{V}(\mathfrak{b}) = \emptyset$.
- (5) Deduce that $f^N \in \mathfrak{a}$ for some $N \geq 1$, and conclude that $\mathbf{I}(\mathbf{V}(\mathfrak{a})) = \sqrt{\mathfrak{a}}$.

Bonus. Show that, conversely, if every maximal ideal of $\mathbb{k}[x]$ is of the form $(x - a)$ with $a \in \mathbb{k}$, then $\mathbb{k}$ is algebraically closed.

Solution. For (1). The quotient

$$K = A/\mathfrak{m}$$

is a field and is finitely generated as a $\mathbb{k}$ -algebra. By Zariski's lemma, K is a finite algebraic extension of $\mathbb{k}$. Since $\mathbb{k}$ is algebraically closed, $K = \mathbb{k}$. Let $a_i \in \mathbb{k}$ be the image of x_i in K . The quotient map is therefore the evaluation homomorphism

$$A \longrightarrow \mathbb{k}, \quad h \longmapsto h(a_1, \dots, a_n),$$

whose kernel is $(x_1 - a_1, \dots, x_n - a_n)$. Hence

$$\mathfrak{m} = (x_1 - a_1, \dots, x_n - a_n).$$

The point a is unique, since the image of each x_i in $A/\mathfrak{m}$ determines a_i .

For (2). A proper ideal $\mathfrak{a}$ is contained in a maximal ideal $\mathfrak{m}$. By (1),

$$\mathfrak{m} = (x_1 - a_1, \dots, x_n - a_n)$$

for some $a \in \mathbb{k}^n$. Every $f \in \mathfrak{a} \subset \mathfrak{m}$ therefore satisfies $f(a) = 0$. Thus every proper ideal has a common zero in $\mathbb{k}^n$.

For (3). Let $h \in \sqrt{\mathfrak{a}}$. Then $h^N \in \mathfrak{a}$ for some $N \geq 1$. If $a \in \mathbf{V}(\mathfrak{a})$, then

$$h(a)^N = 0,$$

so $h(a) = 0$. Hence $h \in \mathbf{I}(\mathbf{V}(\mathfrak{a}))$, proving

$$\sqrt{\mathfrak{a}} \subset \mathbf{I}(\mathbf{V}(\mathfrak{a})).$$

For (4). Suppose that $(a, b) \in \mathbb{k}^{n+1}$ were a common zero of $\mathfrak{b}$. Since every element of $\mathfrak{a}$ vanishes at a , we have $a \in \mathbf{V}(\mathfrak{a})$, hence $f(a) = 0$. But the equation $1 - yf = 0$ would then give

$$1 - bf(a) = 1 = 0,$$

a contradiction. Thus $\mathbf{V}(\mathfrak{b}) = \emptyset$.

For (5). If $\mathfrak{b}$ were proper, (2), applied in the polynomial ring $A[y] = \mathbb{k}[x_1, \dots, x_n, y]$, would give a common zero of $\mathfrak{b}$, contradicting (4). Therefore $\mathfrak{b} = A[y]$, so $1 \in \mathfrak{b}$. Hence there exist $a_1, \dots, a_r \in \mathfrak{a}$ and $g_1, \dots, g_r, h \in A[y]$ such that

$$1 = \sum_{i=1}^r g_i a_i + h(1 - yf).$$

Apply the homomorphism

$$A[y] \longrightarrow A_f, \quad y \longmapsto f^{-1}.$$

The last term vanishes, and we obtain $1 \in \mathfrak{a}A_f$. Clearing denominators gives

$$f^N \in \mathfrak{a}$$

for some $N \geq 1$. Thus $f \in \sqrt{\mathfrak{a}}$, and therefore

$$\mathbf{I}(\mathbf{V}(\mathfrak{a})) \subset \sqrt{\mathfrak{a}}.$$

Together with (3), this proves Hilbert's Nullstellensatz

$$\mathbf{I}(\mathbf{V}(\mathfrak{a})) = \sqrt{\mathfrak{a}}.$$

Bonus. Argue by contraposition. If $\mathbb{k}$ is not algebraically closed, there is an irreducible polynomial $q \in \mathbb{k}[x]$ of degree greater than 1. Then (q) is a maximal ideal of $\mathbb{k}[x]$, but it is not of the form $(x - a)$. Hence the stated property of maximal ideals implies that $\mathbb{k}$ is algebraically closed.

Exercise 3. The affine plane and a punctured affine plane

Set $A = \mathbb{R}[x, y]$ and $X = \mathbb{A}_{\mathbb{R}}^2 = \text{Spec}(A)$. Let $f = x^2 + y^2 - 1$ and, for $\lambda \in \mathbb{R}$, let $g_\lambda = x - \lambda$.

$$C = \text{Spec}(A/(f)), \quad L_\lambda = \text{Spec}(A/(g_\lambda)), \quad Z_\lambda = \text{Spec}(A/(fg_\lambda)).$$

- (1) Show that C and L_λ are the two irreducible components of Z_λ . Determine the scheme $W_\lambda = \text{Spec}(A/(f, g_\lambda))$ and discuss the cases $|\lambda| < 1$, $|\lambda| = 1$, and $|\lambda| > 1$.
- (2) Using lines through the point $(-1, 0) \in C$, give a rational parametrisation of C and deduce that $K(C) \simeq \mathbb{R}(t)$. Determine also $K(L_\lambda)$.

- (3) Let $U = X \setminus \mathbb{W}(x, y)$. Show that $U = D(x) \cup D(y)$ and $\Gamma(\mathcal{O}_U) \simeq \mathbb{R}[x, y]$.
- (4) Describe the affinisisation morphism $U \rightarrow \operatorname{Spec}(\Gamma(\mathcal{O}_U))$ and deduce that U is not affine.
- (5) Let $0 \neq h \in \mathbb{R}[x, y]$ and put $Y_h = X \setminus \mathbb{W}(h)$. Show that Y_h is affine. Is Y_h connected?

Bonus. Is $Y_h(\mathbb{R})$ necessarily connected?

Solution. For (1). The polynomial $x - \lambda$ is irreducible as it has degree 1 in x . The polynomial $x^2 + y^2 - 1$ is also irreducible in $\mathbb{R}[x, y]$: if it factored, since it has degree 2 it would be a product of two linear polynomials, and its homogeneous part of degree 2, namely $x^2 + y^2$, would be a product of two real linear forms. This is impossible, since $x^2 + y^2$ has no non-zero real zero whereas a product of real linear forms does. One could also argue that, if $x^2 + y^2 - 1$ factored as uv , a product of linear forms, then the affine algebraic subset $\mathbb{V}(x^2 + y^2 - 1) \subset \mathbb{R}^2$ (the unit circle) would be the union of the two lines $\mathbb{V}(u)$ and $\mathbb{V}(v)$, which is manifestly not the case.

Hence (f) and (g_λ) are prime ideals, and C and L_λ are irreducible. Their union is Z_λ , so C and L_λ are the irreducible components of Z_λ . For this last statement, one can use the lemma that says: if a topological space is writtten as $X = \cup_{1 \leq i \leq s} X_i$ with $X_1, \dots, X_s$ closed and irreducible, not contained in one another, then $X_1, \dots, X_s$ are the irreducible components of X .

The coordinate ring of W_λ is

$$\frac{\mathbb{R}[x, y]}{(x^2 + y^2 - 1, x - \lambda)} \simeq \frac{\mathbb{R}[y]}{(y^2 + \lambda^2 - 1)}.$$

If $|\lambda| < 1$, put $r = \sqrt{1 - \lambda^2} > 0$. Then

$$y^2 + \lambda^2 - 1 = (y - r)(y + r),$$

and the two factors are coprime, so

$$W_\lambda \simeq \operatorname{Spec}(\mathbb{R} \times \mathbb{R}).$$

Thus W_λ consists of two distinct reduced $\mathbb{R}$ -rational points. If $|\lambda| = 1$, then

$$W_\lambda \simeq \operatorname{Spec}(\mathbb{R}[y]/(y^2)),$$

a single non-reduced $\mathbb{R}$ -rational point. Finally, if $|\lambda| > 1$, put $r = \sqrt{\lambda^2 - 1} > 0$. Then $y^2 + r^2$ is irreducible over $\mathbb{R}$, and

$$W_\lambda \simeq \operatorname{Spec}(\mathbb{C}).$$

Thus in this case C and L_λ have no common real rational point, but W_λ is one closed point with residue field $\mathbb{C}$.

For (2). A line through $(-1, 0)$ with slope t has equation

$$y = t(x + 1).$$

Substituting in $x^2 + y^2 = 1$ gives

$$(x + 1)((1 + t^2)x + t^2 - 1) = 0.$$

Besides the fixed point $(-1, 0)$, the second intersection point is therefore

$$x = \frac{1 - t^2}{1 + t^2}, \quad y = \frac{2t}{1 + t^2}.$$

Conversely, away from $(-1, 0)$ the slope is

$$t = \frac{y}{x + 1}.$$

This gives an isomorphism

$$\operatorname{Spec}(\mathbb{R}[t]) \supset D(1 + t^2) \rightarrow D(1 + x) \subset C,$$

whose inverse is induced by $t \mapsto y/(1+x)$. Thus this is a birational parametrisation, which gives

$$K(C) \simeq \mathbb{R}(t).$$

Moreover $L_\lambda \simeq \mathbb{A}_{\mathbb{R}}^1$ with coordinate y , hence

$$K(L_\lambda) = \mathbb{R}(y) \simeq \mathbb{R}(t).$$

For (3). Since

$$X \setminus \mathbb{W}(x, y) = D(x) \cup D(y),$$

the two affine opens have rings of sections

$$\Gamma(\mathcal{O}_{D(x)}) = \mathbb{R}[x, y]_x, \quad \Gamma(\mathcal{O}_{D(y)}) = \mathbb{R}[x, y]_y,$$

and their intersection is $D(xy)$, with ring of sections $\mathbb{R}[x, y]_{xy}$. The sheaf condition identifies $\Gamma(\mathcal{O}_U)$ with

$$\mathbb{R}[x, y]_x \cap \mathbb{R}[x, y]_y \subset \mathbb{R}(x, y).$$

If $h = a/x^m = b/y^n$ belongs to this intersection, then $y^n a = x^m b$. Since x and y are relatively prime in the UFD $\mathbb{R}[x, y]$, the divisibility $x^m \mid y^n a$ implies $x^m \mid a$, hence $h \in \mathbb{R}[x, y]$. The reverse inclusion is immediate, so

$$\Gamma(\mathcal{O}_U) = \mathbb{R}[x, y].$$

For (4). By (3), the affinisation morphism is

$$U \longrightarrow \operatorname{Spec}(\mathbb{R}[x, y]) = X.$$

The map on global sections is the identity of $\mathbb{R}[x, y]$, so this is precisely the open immersion $U \hookrightarrow X$. It misses the closed point (x, y) and is therefore not an isomorphism. If U were affine, its canonical morphism to $\operatorname{Spec}(\Gamma(\mathcal{O}_U))$ would be an isomorphism. Hence U is not affine.

For (5). For $0 \neq h \in \mathbb{R}[x, y]$,

$$Y_h = X \setminus \mathbb{W}(h) = D(h) \simeq \operatorname{Spec}(\mathbb{R}[x, y]_h),$$

so it is affine. It is also connected. Indeed X is irreducible because $\mathbb{R}[x, y]$ is a domain, and $D(h)$ is a non-empty open subset: the generic point (0) belongs to $D(h)$ because $h \neq 0$. Every non-empty open subset of an irreducible space is irreducible, hence connected.

Bonus. No. For instance, if $h = x^2 + y^2$, then

$$Y_h(\mathbb{R}) = \mathbb{R}^2 \setminus \{(0, 0)\},$$

which is connected. On the other hand, if $h = x$, then

$$Y_h(\mathbb{R}) = \{(x, y) \in \mathbb{R}^2 \mid x \neq 0\},$$

which is the disjoint union of the two non-empty open subsets $x > 0$ and $x < 0$, and is therefore disconnected.

In the last bonus question, connectedness of $Y_h(\mathbb{R})$ is understood with respect to the usual Euclidean topology on $\mathbb{R}^2$.